Experiment 13: Host Discovery in Kali Linux

Aim

To perform host discovery in Kali Linux using Nmap to identify active hosts on a network.

Algorithm

1. Start Kali Linux and open the terminal.
2. Check whether Nmap is installed.
3. Execute a host discovery command to check whether the target host is active.
4. Perform a ping scan to discover active hosts on an authorized local network.
5. Observe and record the results displayed in the terminal.

Commands and Uses

1. Check whether a host is active

nmap -sn 127.0.0.1

Use: Checks host availability without performing a port scan.

2. Discover hosts on a local network

nmap -sn 192.168.1.0/24

Use: Discovers responding hosts in the 192.168.1.0/24 network. Replace this network with your actual local subnet.

3. Check a specific host

ping -c 4 127.0.0.1

Use: Sends four ICMP echo requests to check connectivity with the local machine.

Sample Output

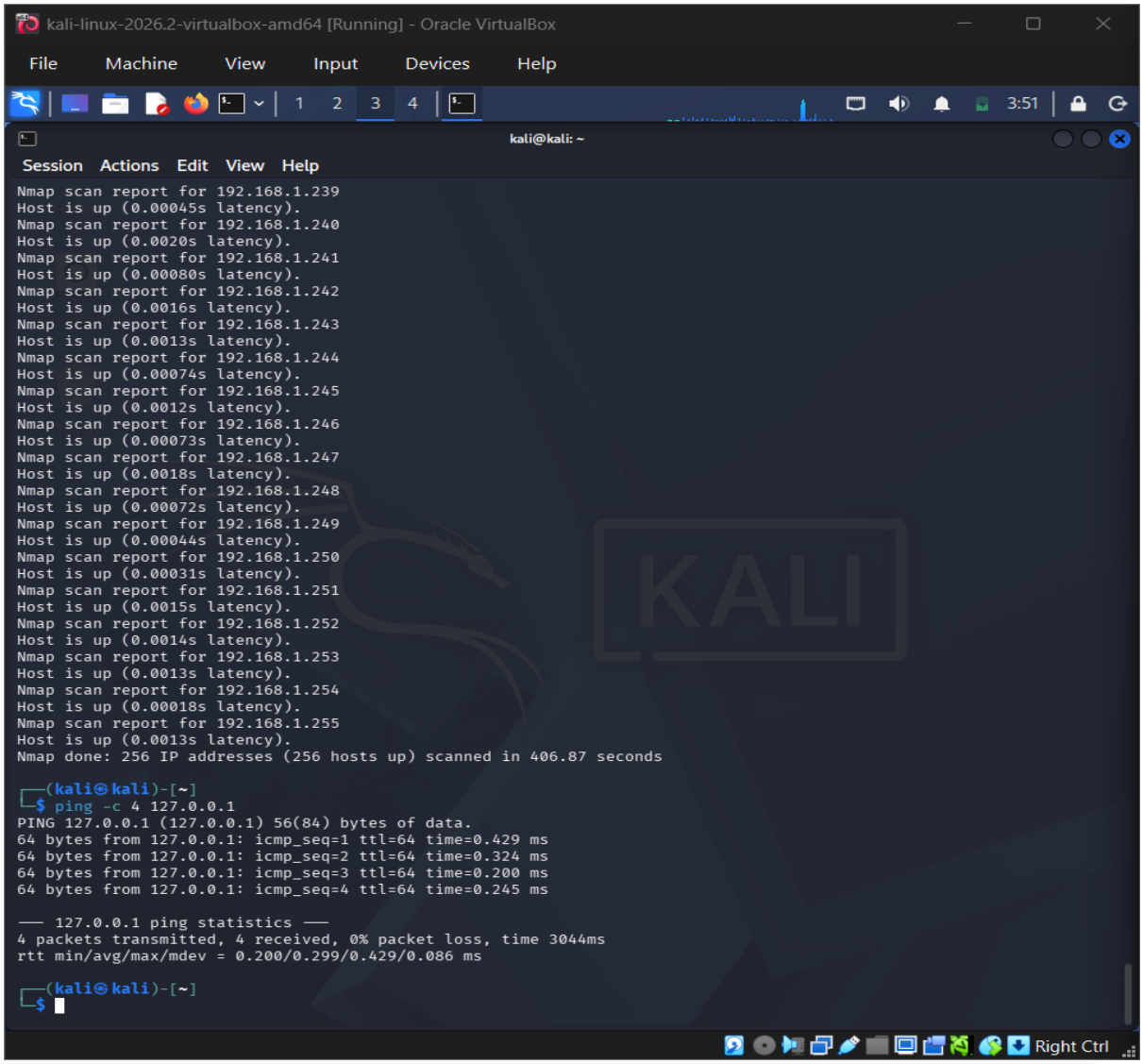
Starting Nmap

Nmap scan report for localhost (127.0.0.1)

Host is up.

Nmap done: 1 IP address scanned

Note: This is illustrative output. Actual results depend on the target and network configuration.



Result

Host discovery was demonstrated using Nmap in Kali Linux to identify active hosts on a network.

Safety: Perform scans only on your own network or systems for which you have permission.